CHE 201 Fall 2019 Quiz 2

Joey Wilczynski

TOTAL POINTS

50 / 100

QUESTION 1

1 85 pts

1.1 a **35 / 40**

- **- 0 pts** Correct
- ✓ **- 5 pts h2 is not calculated or wrong**
- **- 10 pts** KE is not calculated
- **- 5 pts** h1 is wrong
- **- 10 pts** PE not calculated

1.2 b+c **0 / 45**

- **- 0 pts** Correct
- ✓ **- 45 pts not done or wrong**
- **- 5 pts** unit conversion
- **- 10 pts** b is wrong

QUESTION 2

2 2 **15 / 15**

- ✓ **- 0 pts Correct**
- **- 15 pts** not done

CHE 201 INTRO TO THERMODYNAMICS **October 25th, 2019**

NAME: Joey Wilczynski **UIN:** ____________________

QUIZ 2: (100 pts) (Close book/closed notes) Show all your work for full credit

1.(85 points) The power output of steam turbine is 5MW, and the inlet and the exit conditions are as follow: P_1=2MPa, T_1=400°C, V_1=50m/s, z_1= 10m; P_2=15kPa, x_2=0.90, V_2=180m/s, z_2=6m. a)Compare the magnitudes of delta h, delta KE, delta PE. b) Determine the work done per unit mass of the steam flowing through the turbine. c) Calculate the mass flow rate of the steam.

2. (15 points) Briefly explain the process that Lanzatech patented. (Monday's guest speaker; Rachel Brenc)

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i\right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e\right)$$

$$\dot{m} = \frac{AV}{v}$$

1 joule= 1 N.m 1 N= 1 kg.m/s^2

1)

Scematic

P_1 = 2MPa
T_1 - 400°C
V_1 = 50 m/s
z_1 = 10 m

P_2 = 15 kPa
x_2 = .90
V_2 = 180 m/s
z_2 = 6 m

Engineering model
- control volume
$\dot{m}_i = \dot{m}_e = \dot{m}$
- steady state
- 1 inlet 1 outlet

Energy Balance : $\dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i \left(h_i - h_e\right) + \left(\frac{V_i^2 - V_e^2}{2}\right)\left(gz_i - gz_e\right)$

Mass Balance : $\dot{m} = \frac{AV}{v}$

a) $(h_i - h_e)$ = ?

(3247.6 - 2799.5)

.1512
20 Bar

ΔH = 448.1

$\left(\frac{V_i^2 - V_e^2}{2}\right)$

$\frac{50^2 - 180^2}{2} = \frac{-14950}{1000} = -14.95$

$(gz_i - gz_e)$
10m - 6m = 4 m

B) $\dot{m}(h_i - h_e)$
3247 - 2799.5

$v = (1 - .90) \cdot V_g + V_f(.90)$

.1

#2 ON Back

2.) The process that was discussed in class was how her company is taking waste gases / fuel and using a Micro organism to break it down, then it gets converted to ethenol to be used for in making things such as Jet fuel clean reuseable energy.